

# Chemical Bonding

## Ionic Bonding

An ionic bond forms through the electrostatic attraction between a positively charged cation and a negatively charged anion, typically formed when a metal transfers one or more electrons to a nonmetal.

## Covalent Bonding

A covalent bond forms when two atoms share one or more pairs of electrons, typically occurring between nonmetal atoms.

A polar covalent bond occurs when electrons are shared unequally between two atoms due to a difference in electronegativity, resulting in a partial positive charge on one atom and a partial negative charge on the other.

## Molecular Geometry

The VSEPR (Valence Shell Electron Pair Repulsion) theory predicts molecular geometry by assuming that electron pairs around a central atom arrange themselves to minimize repulsion, for example a tetrahedral geometry for four bonding pairs with no lone pairs.